Gabriel Vance
DSC680 – Applied Data Science
Professor Amirfarrokh Iranitalab
January 25, 2026

Project 2

Milestone 1

The S&P 500 is composed of the 500 largest publicly traded companies in the United States and is widely regarded as a benchmark for American economic performance and investment wealth. While long-term growth is not guaranteed, the index has historically trended upward over time. Investors, hedge funds, and analysts continuously attempt to predict both the direction and magnitude of market movements, often with limited success. A wide range of factors—such as weather patterns, global events, monetary policy decisions by the Federal Reserve, and broader macroeconomic conditions—have been examined for potential correlations with market behavior. As a result, financial markets remain the subject of constant analysis in the search for predictive signals.

American politics generally operates on a two-year electoral cycle, with national elections held every two years. Presidential elections occur every four years, with midterm elections taking place in between. This raises an important question: what relevance, if any, do these political cycles have on stock market performance? Do returns in the S&P 500 differ depending on which political party is in power? Additionally, does market performance vary when the presidency and Congress are controlled by the same party versus divided government?

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This project explores the relationship between politics, elections, and the stock market. Specifically, it seeks to answer the following questions:

1. Are future S&P 500 returns statistically similar under both major political parties?
2. Are incumbent political parties more likely to lose elections during periods of S&P 500 underperformance?
3. Given the current political landscape, what is the predicted stock market return from January 2026 to January 2027?

To conduct this analysis, we will use the *yfinance* API to obtain historical S&P 500 prices and annual returns. Political party control and changes in power will be sourced from Kaggle datasets and supplemented with additional historical records. Using this combined dataset, we will apply linear regression models trained on past market and election data to estimate future percentage changes in the stock market. Finally, the project will address ethical considerations related to market prediction, including insider trading concerns, the distinction between probability and expectation, and the broader responsibility associated with providing financial insights in predictive markets.